

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

I J. Kirthiga / 192421265 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Kirthiga
Signature of the student:

19/20
29/11/26

Annexure A

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range** of host IP addresses.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. SUBNET DESIGN FOR SMART CAMPUS NETWORK

The given IP block is 192.168.100.0/24. Using VLSM (Variable Length Subnet Masking), subnets are allocated based on departmental requirements, starting from the largest to smallest.

Given:

- Base Network: 192.168.100.0/24 -256 total IPs
- Subnets required:
 - Computing → /25
 - Electronics → /26
 - Mechanical → /27

Step 1: Allocate Subnets (Largest → Smallest)

We assign from largest subnet to smallest:

Department	Subnet Mask	Size
Computing	/25	128 IPs
Electronics	/26	64 IPs
Mechanical	/27	32 IPs

Step 2: Subnet Details

School of Computing (/25)

- Subnet: 192.168.100.0/25
- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127

- Valid Host Range:
192.168.100.1 → 192.168.100.126
- Usable Hosts:
 $128 - 2 = 126$ hosts

School of Electronics (/26)

- Subnet: 192.168.100.128/26
- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191
- Valid Host Range:
192.168.100.129 → 192.168.100.190
- Usable Hosts:
 $64 - 2 = 62$ hosts

School of Mechanical Engineering (/27)

- Subnet: 192.168.100.192/27
- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range:
192.168.100.193 → 192.168.100.222
- Usable Hosts:
 $32 - 2 = 30$ hosts

Step 3: Remaining Unused IP Range

After allocation:

- Used up to: 192.168.100.223
- Remaining block:

Unused Range:
• 192.168.100.224
This is:

Unused Range:

- 192.168.100.224 → 192.168.100.255

This is:

- 32 IPs remaining
- Can be used as another /27 subnet in future

Department	Network	Host Range	Broadcast	Usable Hosts
Computing	192.168.100.0/25	1 – 126	127	126
Electronics	192.168.100.128/26	129 – 190	191	62
Mechanical	192.168.100.192/27	193 – 222	223	30
Unused	192.168.100.224/27	225 – 254	255	30

The network has been efficiently divided using VLSM, ensuring:

- Optimal utilization of IP addresses
- Sufficient host allocation for each department
- Availability of an extra subnet for future expansion

2. Given:

A company is assigned the IP block 10.10.0.0/16. Different departments require different subnet sizes.

Subnet masks:

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Using VLSM (Variable Length Subnet Masking), subnets are allocated from largest to smallest to ensure efficient utilization of IP addresses

Step 1: Subnet Allocation

1. Software Development (/24)

- Network Address: 10.10.0.0
- Broadcast Address: 10.10.0.255
- Total IPs: 256

2. Quality Assurance (/25)

- Network Address: 10.10.1.0
- Broadcast Address: 10.10.1.127
- Total IPs: 128

3. Human Resources (/26)

- Network Address: 10.10.1.128
- Broadcast Address: 10.10.1.191
- Total IPs: 64

4. Executive Management (/27)

- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Total IPs: 32

Step 2: Valid Host Ranges

- Software Development:
10.10.0.1 – 10.10.0.254
- Quality Assurance:
10.10.1.1 – 10.10.1.126
- Human Resources:
10.10.1.129 – 10.10.1.190
- Executive Management:
10.10.1.193 – 10.10.1.222

Step 3: Number of Usable Hosts

Using formula: $2^n - 2$

- /24 → 254 usable hosts
- /25 → 126 usable hosts
- /26 → 62 usable hosts
- /27 → 30 usable hosts

Step 4: Unused IP Address Range

Allocated range used:
10.10.0.0 to 10.10.1.223

Remaining unused range:
10.10.1.224 to 10.10.255.255

This unused portion can be further subnetted for future departments or expansion.

Conclusion

The subnetting is performed using VLSM, allocating address space from largest to smallest subnet. This approach ensures:

- Efficient utilization of IP addresses
- No overlap between subnets
- Reduced wastage of address space
- Scalability for future network expansion

3. Given

IPv6 Address: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed IPv6 Notation

Rules:

- Remove leading zeros in each block
- Replace consecutive 0000 blocks with :: (only once)

Compressed Address:

2001:db8:abcd::/64

2. Expanded (Full) IPv6 Address

Rewriting the compressed form back to full 8 blocks:

Expanded Address:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix (/64)

- IPv6 uses first 64 bits for network
- Remaining 64 bits for host/interface

Network Prefix:

2001:0db8:abcd:0000::/64

This means:

- First 4 blocks = Network portion
- Last 4 blocks = Host portion

4. Number of Host Addresses in /64

Formula:

$$2^{(128-64)} = 2^{64}$$

Total host addresses:

$$2^{64} = 18,446,744,073,709,551,61$$

5. Total Possible Host Addresses

In IPv6:

- No concept of "usable hosts minus 2" like IPv4
- All addresses in subnet are usable (except reserved cases)

$$\text{Total possible hosts} = 2^{64}$$

= 18.4 quintillion addresses

Conclusion

The IPv6 /64 subnet provides an extremely large address space, making it ideal for smart hospital environments with thousands of IoT devices. The use of IPv6 ensures scalability, efficient routing, and future-proof network design.

4. Given:

Routing Table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP Address: 192.168.2.45

1. Destination Subnet

Subnet mask for /24 = 255.255.255.0

Check the network:

- IP: 192.168.2.45
- Network portion: 192.168.2.0

Destination subnet: 192.168.2.0/24

2. Matching Routing Table Entry

Compare destination with routing table:

- 192.168.1.0/24 (not matching)
- 192.168.2.0/24 (matches)
- 192.168.3.0/24 (not matching)

Matching entry: 192.168.2.0/24

3. Next-Hop / Interface

Since the destination matches 192.168.2.0/24, the router forwards the packet:

Through the interface connected to 192.168.2.0/24 network

(or)

To the next-hop router configured for this subnet

In most cases, this will be the directly connected interface of the router.

4. Default Route (if no match)

If no routing table entry matches:

Router uses Default Route: 0.0.0.0/0

- This forwards packets to the ISP or gateway router
- Used for unknown or external networks

Conclusion

The router performs longest prefix matching and correctly identifies that the destination IP 192.168.2.45 belongs to the 192.168.2.0/24 subnet, forwarding it through the corresponding interface. If no match existed, the packet would be routed using the default route (0.0.0.0/0).

5. Administration LAN → 192.168.10.0/26

- Finance LAN → 192.168.10.64/26

Router Interfaces:

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

1. Network Address & Broadcast Address

Administration LAN (/26)

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63

Finance LAN (/26)

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127

2. Valid Host IP Range

Administration LAN

- 192.168.10.1 – 192.168.10.62

Finance LAN

- 192.168.10.65 – 192.168.10.126

3. Suitable IP Addresses for Computers

Administration LAN (examples)

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Finance LAN (examples)

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67

- PC3 → 192.168.10.68

4. Default Gateway for Each LAN

- Administration LAN Gateway: 192.168.10.1
- Finance LAN Gateway: 192.168.10.65

All hosts in each LAN must use their respective gateway IP to communicate with other networks.

Conclusion

Both LANs are properly subnetted using /26, providing 62 usable hosts per subnet. Each LAN uses a dedicated gateway configured on the router, enabling efficient inter-network communication between Administration and Finance departments.

